

Library Circulation & Digital Resource Database Management System

System Architecture, Frontend Implementation & Backend Database Connectivity

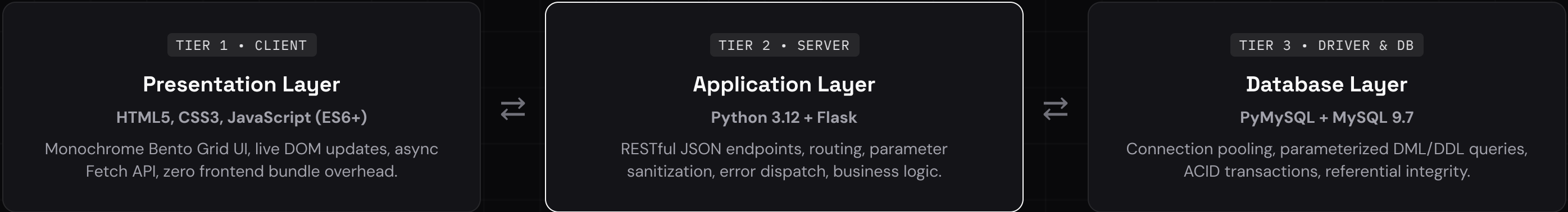
COURSE PROJECT PRESENTATION

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Course: Database Management Systems (DBMS)

End-to-End 3-Tier Enterprise Architecture

Decoupled separation of concerns from Client UI to Relational Database



Modern Client-Side Implementation

Engineered with Vanilla web standards for peak performance, clarity, and zero external bloat

MARKUP

HTML5 Semantic Layout

- Modular structure with Bento Grid layout for catalog cards and quick metrics.
- Interactive modal dialogs for Insert, Delete, and Issue record operations.
- Tabbed navigation for Books, Members, Loans, Digital Resources, and Custom SQL Queries.

STYLING

Vanilla CSS3 Design System

- High-contrast **Monochrome & Dark Mode** palette (#09090B void, #18181B surfaces).
- Fluid CSS Flexbox & CSS Grid layouts with responsive breakpoints.
- Custom micro-interactions, smooth hover transitions, and status pill badges.

CLIENT LOGIC

JavaScript (ES6+)

- `fetch()` asynchronous API calls for seamless real-time data sync without full page reload.
- Dynamic DOM rendering for data tables and search/filtering algorithms.
- Live toast notification banners for operational confirmation (Insert/Delete success).

Server Architecture & Database Connectivity

Robust Python Flask REST microservices backed by MySQL relational database

WEB FRAMEWORK

Python 3.12 & Flask

- Lightweight WSGI framework serving RESTful JSON endpoints.
- Configurable environment via `config.py` for local MySQL instance credentials.
- Centralized error handling returning JSON error payloads with proper HTTP status codes.

DATABASE DRIVER

PyMySQL Connector

- Pure-Python MySQL client driver providing direct TCP socket connection.
- **Parameterized Queries (%s)** across all endpoints to prevent SQL injection vulnerabilities.
- Automatic transaction management with explicit `conn.commit()` and `conn.rollback()` .

RDBMS

MySQL 9.7 Server

- Normalized relational schema (`LibraryDbms`) maintaining 3NF integrity.
- Foreign Key constraints with `ON DELETE CASCADE / RESTRICT` .
- Complex JOIN queries, aggregations, and fine calculation views executed natively.

CRUD Operations & REST API Endpoints

Direct mapping of UI interactions to SQL database mutations

OPERATION: INSERT

Record Creation

- **Endpoint:** `POST /api/books`
- **UI Action:** "Add New Book" Modal with ISBN, Title, Category, Author, Publisher.
- **SQL Execution:** `INSERT INTO BOOK (...)`
`VALUES (...)`
- **Result:** Real-time table refresh + toast notification.

OPERATION: DELETE

Record Removal

- **Endpoint:** `DELETE /api/books/<id>`
- **UI Action:** Delete button on catalog rows with confirmation prompt.
- **SQL Execution:** `DELETE FROM BOOK WHERE`
`BookID = %s`
- **Result:** Row smoothly unmounts; constraints verified.

OPERATION: VIEW

Live Query & Read

- **Endpoints:** `GET /api/books` , `GET /api/issues`
- **UI Action:** Live catalog table, member loan ledger, metrics cards.
- **SQL Execution:** Multi-table `LEFT JOIN` queries for live availability status.
- **Result:** Instant search, sort, and status badges.